

A Survey of Gaussian Splatting and Neural Rendering for Dynamic 3D Scene Reconstruction

InteractiveSurvey

Abstract

The reconstruction of dynamic 3D scenes from multi-view imagery is a fundamental challenge in computer vision, critical for applications ranging from virtual reality to autonomous navigation. While Neural Radiance Fields initially advanced novel view synthesis, the recent emergence of 3D Gaussian Splatting has established a new paradigm by enabling real-time rendering through explicit anisotropic primitives. This survey provides a comprehensive review of methodologies adapting Gaussian Splatting for dynamic environments, specifically addressing the complexities of non-rigid deformations and temporal consistency. We systematically categorize existing approaches into three core themes: spatiotemporal motion modeling, dynamic-static scene decomposition, and efficient representation compression. Our analysis reveals that canonical deformation fields and hierarchical motion anchors effectively decouple geometry from dynamics, while rigidity-aware constraints and learnable 3D masks significantly mitigate artifacts in mixed static-dynamic settings. Furthermore, we demonstrate that sparse anchor-based parameterization and temporal pruning strategies are essential for achieving scalable storage without compromising high-fidelity reconstruction. By critically evaluating the strengths and limitations of current techniques, this work identifies persistent challenges such as temporal jitter and optimization instability. Ultimately, this survey offers a structured taxonomy of the field and outlines promising future directions to enhance robustness and facilitate widespread adoption in resource-constrained applications.

1 Introduction

The reconstruction of dynamic 3D scenes from multi-view imagery represents a cornerstone challenge in computer vision, with profound impli-

cations for virtual reality, autonomous navigation, and digital heritage preservation. Traditional approaches, largely grounded in multi-view stereo and volumetric rendering, have long struggled to balance high-fidelity geometric recovery with the computational efficiency required for real-time applications. The recent advent of Neural Radiance Fields (NeRF) marked a significant paradigm shift by leveraging implicit neural representations to achieve photorealistic novel view synthesis [1]. However, despite their visual quality, NeRF-based methods often incur prohibitive training and rendering costs, limiting their utility in interactive scenarios. This landscape has been further revolutionized by 3D Gaussian Splatting (3DGS), which replaces implicit volumes with explicit, anisotropic Gaussian primitives [2]. By utilizing a rasterization-based rendering pipeline, 3DGS achieves real-time frame rates while maintaining superior image quality, establishing a new state-of-the-art for static scene representation and setting the stage for its extension into the temporal domain [3].

Building upon this foundation, this survey paper provides a comprehensive review of recent advancements in adapting Gaussian Splatting and neural rendering techniques specifically for dynamic 3D scene reconstruction [4]. While the original 3DGS formulation excels in static environments, extending it to handle non-rigid deformations, topological changes, and complex temporal dynamics introduces unique optimization challenges. The primary focus of this work is to categorize and analyze the diverse methodologies proposed to overcome these hurdles, ranging from spatiotemporal motion modeling to efficient compression strategies. We examine how researchers are bridging the gap between the efficiency of explicit Gaussian representations and the flexibility required to model evolving scenes, offering a structured perspective on the current state of 4D

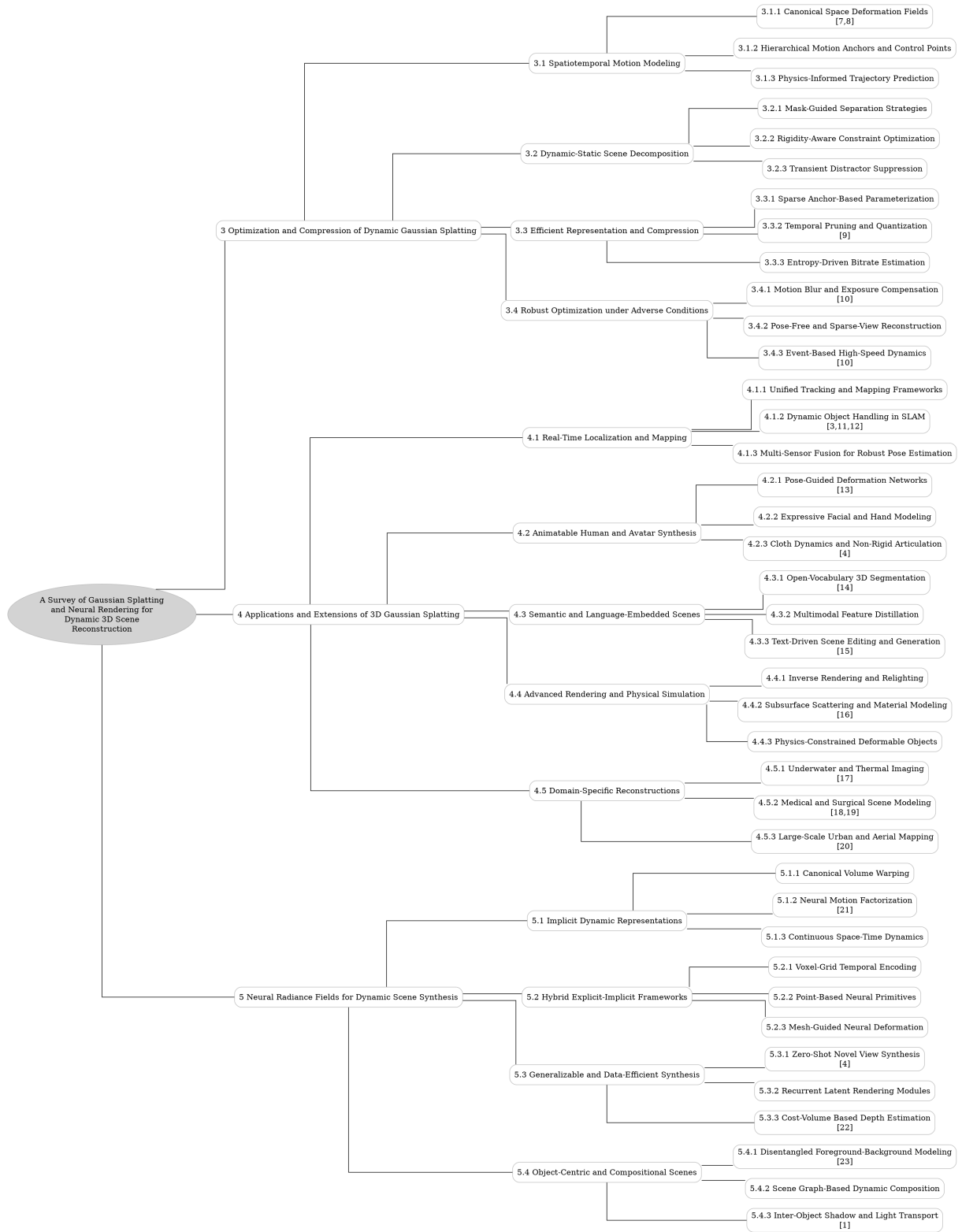


Figure 1: The outline of our survey: A Survey of Gaussian Splatting and Neural Rendering for Dynamic 3D Scene Reconstruction

$$c = \sum_{i=1}^n c_i \alpha_i \prod_{j=1}^{i-1} (1 - \alpha_j)$$

Figure 2: Chart from 'EndoGaussians: Single View Dynamic Gaussian Splatting for Deformable Endoscopic Tissues Reconstruction' (arXiv:2401.13352)



Figure 3: Chart from 'WildGaussians: 3D Gaussian Splatting in the Wild' (arXiv:2407.08447)

$$\hat{\mathcal{I}} = \sum_{k \in \mathcal{G}} c_k \alpha_k \prod_{j=1}^{k-1} (1 - \alpha_j),$$

Figure 4: Chart from 'CaRtGS: Computational Alignment for Real-Time Gaussian Splatting SLAM' (arXiv:2410.00486)

reconstruction.

$$\omega_{ij} = \frac{\exp(-\|\mu_j - x_i\|^2 / \rho_i^2)}{\sum_{x_k \in \mathcal{A}_j} \exp(-\|\mu_j - x_k\|^2 / \rho_k^2)}.$$

Figure 5: Chart from 'HAIF-GS: Hierarchical and Induced Flow-Guided Gaussian Splatting for Dynamic Scene' (arXiv:2506.09518)

The first major theme explored in this survey concerns the mechanisms for spatiotemporal motion modeling, which are essential for decoupling static geometry from temporal dynamics. We analyze approaches that utilize canonical space deformation fields, where a single set of Gaussians is warped over time via learnable neural networks, as well as hierarchical methods employing motion anchors and control points to govern dense clusters. Furthermore, we discuss physics-informed trajectory prediction techniques that integrate explicit motion priors to stabilize optimization and prevent temporal averaging, ensuring that high-frequency details and rapid movements are accurately captured without sacrificing structural consistency.

Subsequently, we delve into strategies for dynamic-static scene decomposition, a critical component for handling scenes containing both moving foregrounds and stationary backgrounds. This section evaluates mask-guided separation strategies that have evolved from relying on external 2D priors to learning adaptive 3D masks jointly with the scene representation. We also examine rigidity-aware constraint optimization, which enforces physical plausibility by decomposing motion into rigid body transformations and non-rigid residuals, and transient distractor suppression techniques that effectively filter out ephemeral objects to prevent artifacts in the reconstructed background. These methods collectively address the ill-posed nature of separating moving elements in unconstrained environments.

Finally, the survey addresses the pressing need for efficient representation and compression to make dynamic Gaussian Splatting scalable for practical deployment. We investigate sparse anchor-based parameterization, which reduces memory footprint by interpolating dense attributes from a compact set of learnable vertices, and temporal pruning and quantization strategies that eliminate redundant primitives across time steps. By analyzing these compression techniques, we highlight how recent works achieve a favorable

trade-off between rendering fidelity and storage efficiency, enabling the reconstruction of high-speed dynamic scenes with minimal computational overhead while preserving the real-time capabilities inherent to the Gaussian splatting formulation [5].

$$\nabla_k = \frac{\sum^N \Psi(k, t) \|\nabla_{k,t}\|}{\sum^N \Psi(k, t)},$$

Figure 6: Chart from 'ADC-GS: Anchor-Driven Deformable and Compressed Gaussian Splatting for Dynamic Scene Reconstruction' (arXiv:2505.08196)

The contributions of this survey are threefold [6]. First, we provide a systematic taxonomy of existing dynamic 3D Gaussian Splatting methods, organizing them by their core methodological innovations in motion modeling, scene decomposition, and compression [7]. Second, we offer a critical analysis of the strengths and limitations of each approach, identifying common failure modes such as temporal jitter, boundary artifacts, and optimization instability. Third, we outline promising future research directions, emphasizing the need for more robust physical priors, better handling of topological changes, and further improvements in compression ratios to facilitate widespread adoption in resource-constrained applications.

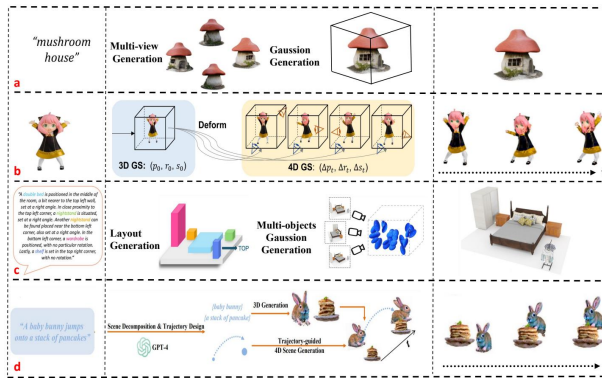


Figure 7: Chart from 'From Fields to Splats: A Cross-Domain Survey of Real-Time Neural Scene Representations' (arXiv:2509.23555)

$$\tilde{b}(t; \theta, \sigma) = \exp \left(-\frac{1}{2\sigma^2} (t - \theta)^2 \right),$$

Figure 8: Chart from 'Deform3DGS: Flexible Deformation for Fast Surgical Scene Reconstruction with Gaussian Splatting' (arXiv:2405.17835)

2 Optimization and Compression of Dynamic Gaussian Splatting

2.1 Spatiotemporal Motion Modeling

2.1.1 Canonical Space Deformation Fields

Canonical space deformation fields constitute a foundational paradigm for extending 3D Gaussian Splatting to dynamic scenes by decoupling static geometry from temporal motion [7]. In this framework, a single set of canonical 3D Gaussians represents the underlying scene structure, while a learnable deformation network predicts time-dependent offsets to transform these primitives into their observed states [8]. This approach effectively maps points from a fixed reference coordinate system to time-specific world coordinates, allowing the model to maintain a compact representation without storing distinct Gaussian parameters for every frame.

The deformation process is typically parameterized by neural networks that ingest spatial coordinates and temporal encodings to regress attributes such as position, rotation, and scale. Advanced implementations often employ a dual-deformation strategy, utilizing separate coarse and fine networks to capture low-frequency global motion and high-frequency local details simultaneously. By conditioning these predictions on learnable embeddings associated with both individual Gaussians and specific timestamps, the system can accurately model complex non-rigid transformations, including subtle surface variations like clothing wrinkles or facial expressions, through implicit field warping.

Despite their efficiency, canonical deformation methods face inherent challenges in handling large displacements and topological changes due to the constraints of a single shared reference space. Optimization difficulties often arise when attempting to maintain consistency across significant temporal gaps, leading to artifacts in regions with rapid motion or occlusion. Consequently, recent advancements have introduced mechanisms such as sliding window strategies and transient-aware aux-

iliary modules to mitigate these limitations, ensuring robust reconstruction fidelity while preserving the real-time rendering capabilities inherent to the Gaussian splatting formulation.

2.1.2 Hierarchical Motion Anchors and Control Points

Hierarchical motion anchors and control points address the computational inefficiency of modeling complex deformations by decoupling scene geometry from motion dynamics. Instead of optimizing individual Gaussian parameters for every frame, these methods utilize a sparse set of learnable anchors to govern the transformation of dense Gaussian clusters. This structural decomposition significantly reduces model capacity requirements while enabling the representation of non-rigid motions through localized deformation fields anchored to specific spatial coordinates.

To capture varying motion complexities, adaptive densification strategies are employed where anchor resolution increases in regions exhibiting high dynamic scores or rapid trajectory changes. Control points are often initialized via farthest point sampling on canonical frames and subsequently refined to minimize rendering loss, ensuring that subtle non-rigid details are preserved without redundant overhead in static areas. By assigning Gaussians to the nearest control point based on trajectory similarity, the system effectively groups primitives with coherent motion patterns, facilitating robust tracking and consistent temporal interpolation.

Furthermore, this hierarchy supports explicit 6-DoF pose estimation for superpoints, allowing for precise motion capture applications where novel view images dictate component movement. The framework mitigates optimization ambiguities inherent in deformation-based approaches by establishing fixed references or flow priors that guide Gaussians to clear destinations across time steps. Consequently, hierarchical anchoring not only enhances the fidelity of dynamic reconstructions in challenging scenarios but also improves training convergence by isolating motion learning from geometric structure maintenance.

2.1.3 Physics-Informed Trajectory Prediction

Physics-informed trajectory prediction in dynamic 3D Gaussian Splatting addresses the limitations of purely data-driven deformation fields by integrating explicit motion priors to stabilize optimization.

Traditional approaches often employ time-variant Multi-Layer Perceptrons to map canonical coordinates and timestamps to spatial offsets, yet these models frequently converge to temporally averaged representations that fail to capture high-frequency or conflicting motion patterns. To mitigate this, recent methodologies decompose scene dynamics by grouping Gaussians into superpoints based on motion similarity, allowing distinct rigid or non-rigid transformations to be optimized per cluster rather than enforcing a global deformation function.

To further enhance physical plausibility, advanced frameworks incorporate regularization terms that enforce spatiotemporal smoothness and piecewise linear trajectory constraints on the predicted scene flow. Instead of directly optimizing raw Gaussian parameters, these systems predict residuals for position, rotation, and scale through lightweight networks conditioned on positional encodings of time and space. This separation of geometric structure from temporal dynamics enables the model to maintain a stable canonical reference while accurately simulating complex behaviors, such as rapid acceleration or periodic oscillation, which are often lost in standard interpolation schemes.

Despite these advancements, balancing computational efficiency with modeling fidelity remains a critical challenge. While introducing hierarchical motion bases or polynomial functions can reduce the parameter count compared to dense neural fields, the inference overhead associated with evaluating time-dependent coefficients often surges in highly dynamic scenes. Consequently, current research trends focus on adaptive strategies that selectively refine trajectories for high-dynamic regions identified via variance metrics, ensuring that computational resources are allocated primarily to areas exhibiting complex motion characteristics without compromising the rendering quality of static backgrounds.

2.2 Dynamic-Static Scene Decomposition

2.2.1 Mask-Guided Separation Strategies

Mask-guided separation strategies have emerged as a pivotal mechanism for disentangling static backgrounds from dynamic foregrounds in 4D scene reconstruction. Early approaches frequently relied on external 2D segmentation priors or manually defined bounding boxes to generate binary

masks, which were subsequently lifted into 3D space to isolate moving Gaussians. While effective in controlled environments, these methods often suffer from boundary artifacts and temporal inconsistencies, particularly when 2D foreground definitions diverge from the true volumetric extent of dynamic regions. Consequently, the reliance on static masks can introduce severe motion blur and detail loss in areas exhibiting complex or rapid deformation, as the rigid separation fails to account for nuanced spatial-temporal variations.

To address these limitations, recent advancements have shifted towards learning dynamic 3D masks that evolve jointly with the scene representation. Instead of fixed priors, these strategies employ learnable parameters or probabilistic classifiers attached to individual Gaussian primitives, enabling the system to automatically distinguish between static and dynamic components based on pixel-change patterns or motion coherence. This adaptive formulation allows for the continuous refinement of separation boundaries during optimization, effectively mitigating the feature collision and over-smoothing issues prevalent in low-rank deformation fields. By integrating consistency losses across temporal windows, these methods ensure that the separation remains robust even under sparse viewpoint conditions or significant occlusion.

Furthermore, sophisticated regularization techniques have been introduced to stabilize the training of mask-guided systems. Morphological operations such as erosion and dilation are often applied to mask boundaries to prevent gradient leakage, while decay schedules modulate the influence of dynamic regions to avoid premature convergence to local minima. Unlike traditional pipelines that reset opacity to remove floaters, modern strategies maintain joint spatial-temporal optimization by eliminating such disruptive reinitializations. This holistic approach not only preserves fine geometric details in high-motion areas but also significantly reduces computational overhead compared to heavy 3D optimization routines, establishing a new standard for efficient and high-fidelity dynamic scene synthesis.

2.2.2 Rigidity-Aware Constraint Optimization

Rigidity-aware constraint optimization addresses the inherent ill-posedness of dynamic scene reconstruction by enforcing physical plausibility

through structured motion priors. Unlike unconstrained deformation fields that often suffer from temporal jitter and geometric inconsistencies, this paradigm decomposes scene dynamics into rigid body transformations and residual non-rigid components. By parameterizing motion via $SE(3)$ transformations applied to grouped Gaussian primitives or control points, the optimization landscape is significantly regularized. This approach leverages linear blend skinning mechanisms to ensure that coherent object parts move synchronously, thereby reducing the degrees of freedom required to model complex articulations while preserving high-frequency details through learned residual offsets.

The core of this strategy involves a hybrid objective function that jointly minimizes photometric reconstruction errors and rigidity-specific regularization terms. During training, the framework typically employs a staggered optimization schedule where canonical geometry is first stabilized before unlocking deformation parameters. To prevent overfitting to noise or ambiguous monocular cues, constraints such as cross-frame consistency losses and spectral-aware encodings are integrated. These mechanisms enforce smoothness across temporal boundaries and penalize deviations from expected rigid motion patterns, effectively distinguishing between static background elements and dynamic foreground objects without requiring explicit segmentation masks.

Furthermore, advanced implementations incorporate adaptive pruning and opacity regularization to maintain computational efficiency while adhering to rigidity constraints. By identifying low-dynamic regions and treating them as static, the method avoids redundant computation associated with modeling stationary geometry as deformable. The optimization process dynamically adjusts the weighting between rigid transformation accuracy and non-rigid residual flexibility, ensuring that the model captures significant geometric changes without compromising temporal stability. This balanced approach enables robust reconstruction of scenes exhibiting both large-scale rigid motions and subtle non-rigid deformations within a unified, end-to-end differentiable framework.

2.2.3 Transient Distractor Suppression

Transient distractor suppression in dynamic neural rendering primarily addresses the challenge of separating ephemeral objects, such as moving pedes-

trians or vehicles, from the underlying static scene geometry. Contemporary approaches often leverage uncertainty estimation or dynamic-static decomposition to identify and exclude these non-permanent elements during the optimization process. By modeling the scene as a composition of persistent and transient components, methods can effectively ignore inconsistent observations across multiple views, thereby preventing moving artifacts from corrupting the reconstructed background representation.

To achieve robust separation, recent techniques integrate temporal consistency constraints and attention mechanisms that weigh input frames based on their geometric stability. Strategies include employing temporal opacity fields where primitives exhibiting high variance over time are automatically pruned or assigned negligible density. Furthermore, advanced frameworks utilize event-based cues or noise-aware regularization to detect rapid motion boundaries, ensuring that the optimizer focuses on stable regions while suppressing flickering artifacts caused by fast-moving distractors without requiring explicit semantic masks.

The efficacy of these suppression mechanisms is critical for maintaining high-fidelity novel view synthesis in unconstrained environments. By dynamically filtering out transient interference, these methods significantly reduce motion blur and preserve fine geometric details in the static background. This leads to improved perceptual quality and temporal coherence, allowing for seamless rendering even in scenes with complex, overlapping motions where traditional multi-view consistency assumptions typically fail due to the presence of dynamic outliers.

2.3 Efficient Representation and Compression

2.3.1 Sparse Anchor-Based Parameterization

Sparse anchor-based parameterization addresses the prohibitive memory costs of dense 4D Gaussian representations by decoupling geometric primitives from their temporal attributes. Instead of optimizing independent parameters for every Gaussian at each timestep, this strategy establishes a sparse set of learnable anchor points that serve as spatial control vertices. These anchors encode implicit features within a compact latent space, effectively acting as a bottleneck that regularizes the representation while preserving high-

frequency geometric details necessary for accurate reconstruction.

The core mechanism involves predicting the attributes of dense Gaussian clouds through interpolation from neighboring sparse anchors using a lightweight Multi-Layer Perceptron (MLP). By mapping spatial coordinates and temporal indices to a high-dimensional feature space via positional encoding, the network synthesizes position deformations, scale variations, and rotation offsets dynamically. This explicit-implicit hybrid formulation allows the model to capture complex non-rigid motions without storing redundant per-frame parameters, significantly reducing the overall parameter count compared to direct optimization or dense voxel-grid approaches.

Furthermore, this parameterization facilitates efficient optimization by enabling staggered training schedules where anchor features stabilize before refining the deformation field. The sparse nature of the anchors naturally supports adaptive densification and pruning strategies, ensuring computational resources focus on regions exhibiting significant dynamic activity. Consequently, this approach achieves a favorable trade-off between rendering fidelity and storage efficiency, allowing for real-time inference and scalable reconstruction of high-speed dynamic scenes with minimal overhead.

2.3.2 Temporal Pruning and Quantization

Temporal pruning strategies in dynamic 3D Gaussian Splatting address the inherent redundancy of representing static or slowly moving regions across continuous time steps. Unlike static scene compression, these methods evaluate Gaussian significance through temporal sensitivity scores or motion magnitude, effectively eliminating primitives that contribute minimally to the evolving scene geometry. By removing low-opacity Gaussians or those exhibiting negligible temporal variation, the framework significantly reduces the primitive count without compromising visual fidelity, thereby accelerating both training convergence and real-time rendering speeds [9].

Complementing structural sparsity, attribute pruning targets the parametric overhead by converting specific 4D Gaussians back into static 3D representations. This process identifies parameters, such as scaling or rotation, that remain time-invariant and freezes them, effectively decoupling temporal degrees of freedom where motion is ab-

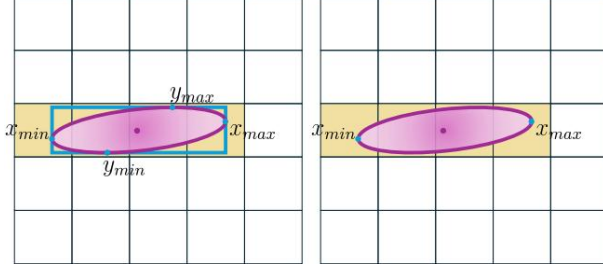


Figure 9: Chart from 'Speedy-Splat: Fast 3D Gaussian Splatting with Sparse Pixels and Sparse Primitives' (arXiv:2412.00578)

sent. Such discriminative pruning not only shrinks the model footprint but also mitigates overfitting to noisy temporal signals, ensuring that the remaining dynamic components focus computational resources on capturing complex, high-frequency motions rather than redundant static backgrounds.

To further enhance storage efficiency, sensitivity-based quantization is applied to the remaining temporal parameters, transforming continuous high-precision values into discrete low-bit representations. Techniques such as vector quantization cluster similar parameter trajectories into codebooks, allowing the system to store indices rather than raw floating-point data. When combined with entropy-based compression, this mixed-precision approach achieves substantial reduction in memory usage while maintaining reconstruction quality, offering a scalable solution for long-sequence dynamic scene modeling without requiring extensive retraining or architectural modifications.

2.3.3 Entropy-Driven Bitrate Estimation

Entropy-driven bitrate estimation leverages statistical redundancy within quantized Gaussian attributes to optimize storage efficiency in dynamic scene representations. By modeling the probability distribution of deformation parameters, such as scale, rotation, and opacity displacements, these methods calculate the theoretical lower bound for bits required per Gaussian. This approach often integrates an entropy regularization term directly into the training loss function, encouraging the model to converge toward sparse, highly predictable attribute distributions that are inherently more compressible without sacrificing visual fidelity.

The estimation process typically distinguishes between static and dynamic regions, assigning

varying bit budgets based on the temporal entropy of specific Gaussian trajectories. In high-motion areas, where prediction error variance increases, the entropy model adapts by allocating higher bitrates to preserve structural details, whereas static regions benefit from aggressive compression due to their low temporal variance. Some frameworks further refine this by utilizing motion segmentation masks to isolate transient elements, ensuring that the entropy coder focuses computational resources on significant geometric changes rather than redundant background information.

Ultimately, this strategy facilitates a unified compression pipeline where bitrate control is intrinsically linked to the scene's information content. By combining entropy coding with vector quantization and structured spatial organization, modern approaches achieve substantial reduction in model size while maintaining real-time rendering performance. The synergy between entropy-based regularization during optimization and subsequent arithmetic coding ensures that the final bitstream closely approximates the theoretical rate-distortion limit, enabling efficient streaming of complex free-viewpoint videos over bandwidth-constrained networks.

2.4 Robust Optimization under Adverse Conditions

2.4.1 Motion Blur and Exposure Compensation

Motion blur remains a critical bottleneck in dynamic scene reconstruction, particularly when fast object velocities or camera shakes violate the static assumption inherent in many neural rendering frameworks. Traditional deformation-based methods often converge to over-smoothed states in regions of intense motion, resulting in perceptible artifacts and loss of high-frequency details. This degradation stems from the inability of standard optimization objectives to distinguish between genuine geometric complexity and blur induced by exposure integration over time. Consequently, baseline approaches frequently fail to preserve sharp boundaries around rapidly moving limbs or facial expressions, leading to significant fidelity drops in novel view synthesis.

To address these limitations, recent advancements integrate explicit blur formation models and auxiliary sensor data to decouple motion from exposure effects. Techniques such as De-

blur4DGS extend Gaussian Splatting by learning a blur kernel alongside scene geometry, enabling the recovery of sharp structures from blurry monocular inputs. Furthermore, the incorporation of event camera data provides high-temporal-resolution cues that guide the reconstruction process; specific event losses simulate logarithmic brightness changes to penalize inconsistencies between rendered frames and actual motion trajectories [10]. These strategies effectively anchor the optimization to ground truth dynamics, preventing the model from overfitting to smeared pixel values.

$$\mathcal{L}_{\text{event}} = w_{ev} \cdot \left| \frac{\Delta \mathcal{L}}{\|\Delta \mathcal{L}\|_F} - \frac{\mathbf{E}_{gt}}{\|\mathbf{E}_{gt}\|_F} \right|_1,$$

Figure 10: Chart from ‘EVENT-GUIDED 3D GAUSSIAN SPLATTING FOR DYNAMIC HUMAN AND SCENE RECONSTRUCTION’ (arXiv:2509.18566)

Complementing data-driven deblurring, exposure compensation mechanisms utilize temporal photometric consistency and adaptive weighting to stabilize training. By warping neighboring frames via estimated 3D scene flow and comparing them against current observations, these methods enforce coherence while accounting for disocclusions through motion-aware masks. Advanced pipelines also employ adaptive thresholding to separate static and dynamic regions, allowing for targeted refinement of Gaussian parameters in areas prone to flickering or blur. Such approaches ensure that opacity resets and densification strategies do not inadvertently erase learned motion cues, thereby maintaining long-term temporal stability and geometric sharpness across complex sequences.

2.4.2 Pose-Free and Sparse-View Reconstruction

Pose-free and sparse-view reconstruction addresses the ill-posed nature of dynamic scene modeling where explicit camera poses are unavailable or input views are severely limited. Traditional pipelines relying on Structure-from-Motion tools often fail in dynamic contexts due to violated static scene assumptions, leading to inaccurate initialization and convergence issues. To mitigate this, recent approaches integrate pose estimation directly into the optimization loop, jointly refining camera parameters and scene representations. These methods frequently leverage geometric priors, such as

depth maps, or enforce photometric consistency across temporal sequences to resolve depth ambiguities inherent in monocular settings.

In sparse-view scenarios, the scarcity of geometric cues exacerbates reconstruction instability, often resulting in artifacts or overfitting. Advanced techniques counteract this by introducing robust regularization schemes that balance data fidelity with motion smoothness and structural coherence. Instead of maintaining independent per-frame representations, which incur high memory costs, modern frameworks utilize canonical spaces augmented with deformation fields or spatio-temporal embeddings. This strategy allows the model to share geometric information across time and views, effectively propagating constraints from observed frames to unobserved regions while maintaining a compact model footprint.

Furthermore, specific architectural innovations enhance robustness under extreme sparsity and pose uncertainty. Methods employing motion-aware grouping cluster primitives with similar trajectories to predict rigid transformations collectively, reducing the degrees of freedom and stabilizing optimization. Additionally, adaptive pruning mechanisms with annealing schedules are utilized to eliminate redundant primitives while tolerating imprecise pose estimates during early training stages. By combining these strategies, current state-of-the-art approaches achieve temporally stable and geometrically plausible reconstructions even when confronted with limited baselines, low-light conditions, or completely uncalibrated monocular video inputs.

2.4.3 Event-Based High-Speed Dynamics

Event-based sensors offer a paradigm shift for modeling high-speed dynamics by capturing asynchronous brightness changes with microsecond latency, effectively bypassing the motion blur inherent in standard frame-based cameras [10]. In the context of 3D Gaussian Splatting, integrating event streams allows for the precise reconstruction of rapid trajectories that traditional discrete sampling fails to resolve. By leveraging the high temporal resolution of event data, recent approaches simulate logarithmic brightness changes between rendered frames to construct specialized loss functions, ensuring that fast-moving primitives align accurately with observed transient signals without suffering from temporal aliasing [10].

To address the computational surge associated with modeling complex motions via time-variant Multi-Layer Perceptrons, efficient strategies employ selective dynamic modeling. This involves partitioning Gaussians based on dynamic scores, treating low-dynamic regions as static to eliminate redundant calculations while dedicating expressive capacity to high-velocity areas. Techniques such as keyframe-based temporal interpolation and polynomial trajectory functions further reduce memory footprints by avoiding the storage of dense information for every timestamp. This decomposition balances the expressiveness required for intricate deformation patterns with the practical constraints of real-time rendering and storage efficiency.

Despite these advancements, maintaining temporal smoothness across partition boundaries remains a critical challenge, often manifesting as visual discontinuities or floating artifacts during ultra-fast motion. To mitigate these issues, cross-frame consistency losses are employed to enforce coherence between adjacent timesteps, while robust initialization protocols prevent velocity-induced artifacts. Furthermore, hybrid frameworks increasingly combine event-driven constraints with scene flow estimation to distinguish between rigid body movements and deformable dynamics. These integrated methods ensure that high-frequency trajectory details are captured faithfully while preserving spatial coherence, marking a significant step toward robust, real-time 4D reconstruction in challenging high-speed scenarios.

3 Applications and Extensions of 3D Gaussian Splatting

3.1 Real-Time Localization and Mapping

3.1.1 Unified Tracking and Mapping Frameworks

Unified tracking and mapping frameworks increasingly leverage explicit 3D Gaussian representations to achieve tight coupling between camera pose estimation and dense scene reconstruction. Unlike traditional methods that treat tracking and mapping as loosely connected modules, modern approaches integrate these processes into a single optimization loop where geometric verification and regularization are performed directly on the Gaussian primitives. This explicit nature allows for robust handling of ambiguities in incremen-

tal reconstruction, enabling state-of-the-art performance in novel view synthesis and trajectory estimation even for intricate details like transparent objects.

To address dynamic environments, these frameworks often employ deformation fields or persistent Gaussian trajectories that map time-varying observations to a canonical space. By modeling scene dynamics through continuous 4D space-time volumes or per-point tracking, systems can maintain temporal consistency without relying on offline structure-from-motion poses. Advanced implementations utilize multi-level positional encoding within neural networks to predict deformations, ensuring that appearance parameters remain constant while allowing means, rotations, and scales to evolve, thereby effectively capturing non-rigid motions and complex topology changes.

Despite these advancements, significant challenges persist in extremely sparse viewpoints and highly dynamic scenarios where performance degrades compared to specialized NeRF-based methods. Current bounded encodings often struggle with large-scale outdoor variations, and the computational cost of physically based methods remains prohibitive for real-time applications in massive environments. Future research directions aim to resolve scale ambiguities in monocular settings, improve robustness against rapid motion through better initialization strategies, and extend these unified frameworks to support lifelong localization in changing environments without sacrificing geometric fidelity.

3.1.2 Dynamic Object Handling in SLAM

Traditional semantic SLAM systems predominantly address dynamic objects by identifying and feature points associated with moving entities during the front-end tracking phase. This strategy relies heavily on pixel-level segmentation to prevent erroneous data associations that could degrade pose estimation accuracy. By filtering out dynamic features before backend optimization, these methods ensure the construction of a consistent static map, effectively treating moving objects as outliers or noise rather than integral scene components. While robust for localization, this exclusionary approach inherently limits the system’s ability to represent or interact with the dynamic elements of the environment.

In contrast, emerging 3D Gaussian Splatting (3DGS) based SLAM frameworks are shifting to-

wards explicit modeling of dynamic primitives rather than their removal [3]. These advanced techniques decompose the scene into static backgrounds and foreground objects, often employing dedicated motion models to estimate the 6DoF transformation of individual entities over time. By allowing Gaussian primitives to persist and move coherently across frames while maintaining consistent appearance parameters, these systems achieve a unified representation that captures both geometry and temporal dynamics [11]. This paradigm enables the simultaneous optimization of camera trajectory and object motion, facilitating accurate tracking even in complex, cluttered environments.

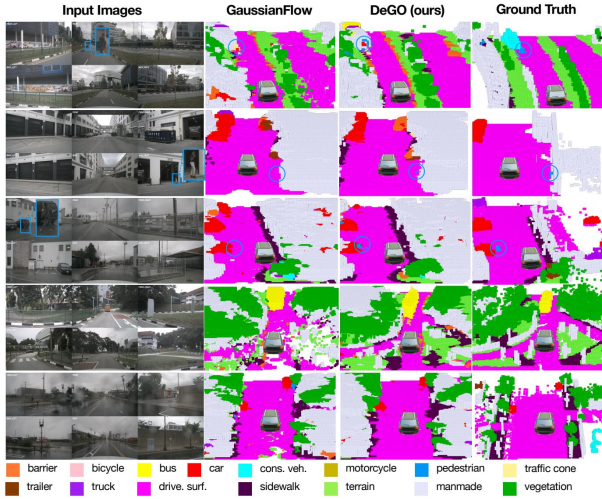


Figure 11: Chart from 'Deformable Gaussian Occupancy: Decoupling Rigid and Nonrigid Motion with Factorized Distillation' (arXiv:2605.28587)

Furthermore, recent developments integrate deformation fields and spatiotemporal encoders to handle non-rigid motions and intricate interactions without relying on predefined object priors. By lifting the representation into a 4D space or utilizing canonical spaces mapped via deformation networks, these methods address the challenges of storage efficiency and temporal consistency inherent in monocular settings. This evolution allows SLAM systems to not only localize within changing environments but also reconstruct high-fidelity dynamic scenes in real-time, supporting downstream applications such as robotic manipulation and augmented reality where understanding object behavior is as critical as knowing camera pose [12].

$$\mathcal{C}_{u,v} = \sum_{i=1}^N \vec{c}_i s_i \cdot \alpha_i \prod_{j=1}^{i-1} (1 - s_j \cdot \alpha_j).$$

Figure 12: Chart from 'DAS3R: Dynamics-Aware Gaussian Splatting for Static Scene Reconstruction' (arXiv:2412.19584)

3.1.3 Multi-Sensor Fusion for Robust Pose Estimation

Multi-sensor fusion has emerged as a critical paradigm for achieving robust 6-DoF pose estimation in dynamic environments where single-modality sensors frequently fail. By integrating complementary data streams, such as LiDAR depth measurements and RGB visual features, these systems effectively mitigate the limitations inherent to RGB-only approaches, including scale ambiguity, drift, and sensitivity to low-texture regions. The tight coupling of heterogeneous sensor inputs allows for the maintenance of accurate tracking trajectories even under rapid motion or significant illumination changes, which often degrade traditional visual odometry performance.

Recent advancements leverage differentiable rendering frameworks to jointly optimize camera poses and scene representations, moving beyond loosely coupled traditional SLAM pipelines. Methods utilizing 3D Gaussian Splatting and Neural Radiance Fields now incorporate multi-modal constraints directly into the optimization loop, enabling the simultaneous refinement of geometric structures and temporal dynamics. This joint optimization strategy significantly enhances alignment accuracy across frames by resolving the ambiguity between object motion and camera ego-motion, particularly in scenarios involving articulated objects or non-rigid deformations where static scene assumptions break down.

Despite these improvements, challenges persist in unbounded outdoor settings and environments with sparse sensor coverage. Current bounded encoding strategies often struggle to capture large-scale variations, while the calibration complexity of multi-sensor rigs can introduce systematic errors if not meticulously managed. Future research directions aim to develop more adaptive fusion architectures that reduce reliance on precise extrinsic calibration and improve generalization to diverse, unstructured scenes. Enhancing the robustness of pose initialization and maintaining consis-

tency in long-term operation remain pivotal for deploying these systems in real-world robotic navigation and augmented reality applications.

3.2 Animatable Human and Avatar Synthesis

3.2.1 Pose-Guided Deformation Networks

Pose-guided deformation networks address the limitations of static representations by explicitly modeling non-rigid dynamics through learned mappings from a canonical space to observation space. Unlike methods relying on simplified motion assumptions or lacking spatial-temporal autocorrelation, these architectures utilize pose parameters as primary conditioning signals to drive geometric transformations. By decoupling appearance from motion, such frameworks effectively handle complex articulated structures and soft tissue deformations, ensuring robust reconstruction even under sparse view conditions where traditional optimization often fails.

The core mechanism typically involves a hierarchical pipeline where a pose embedding network extracts high-level kinematic features from coarse mesh estimates or skeletal data. These embeddings condition subsequent deformation modules, such as Relative Mesh Deformation and Relative Gaussian Deformation networks, which predict localized offsets for position, rotation, and scaling parameters. This design allows for the precise refinement of dynamic meshes while simultaneously adjusting associated 3D Gaussian primitives, facilitating a hybrid strategy that combines linear blend skinning for rigid body movements with neural fields for secondary non-rigid effects like muscle jiggles.

Advanced implementations further enhance fidelity by employing independent deformation networks for each Gaussian attribute, optimized via photometric and geometric consistency losses across time. This parameter-specific approach prevents the entanglement of motion and appearance, enabling the system to capture fine-grained temporal residuals without requiring explicit temporal cues during inference. Consequently, pose-guided networks achieve superior rendering quality and generalization to novel poses compared to global deformation fields, offering a scalable solution for real-time dynamic novel view synthesis in unconstrained environments [13].

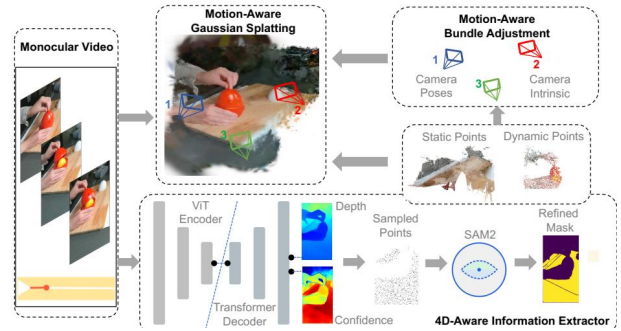


Figure 13: Chart from '4D3R: Motion-Aware Neural Reconstruction and Rendering of Dynamic Scenes from Monocular Videos' (arXiv:2511.05229)

3.2.2 Expressive Facial and Hand Modeling

Recent advancements in expressive facial and hand modeling have shifted from rigid parametric templates to hybrid frameworks that integrate morphable models with deformable 3D Gaussian primitives. By coupling established priors such as SMPL and FLAME with learnable deformation fields, contemporary methods effectively disentangle rigid pose transformations from non-rigid micro-expressions. This synthesis allows for the precise capture of high-frequency geometric details, including skin wrinkles and intricate finger articulations, which traditional mesh-based approaches often smooth over due to topological constraints.

To address the challenge of topology variation and complex occlusions inherent in hand gestures and speech-driven animations, researchers have introduced expression-aware deformation modules and latent feature grids. These systems utilize multi-layer perceptrons to predict Gaussian attributes dynamically based on facial action units or hand joint angles, enabling realistic rendering of exaggerated expressions and self-occluding fingers without requiring explicit mesh reconstruction. Furthermore, the adoption of sparse control points and linear blend skinning extensions facilitates efficient coarse-to-fine refinement, ensuring that local deformations remain consistent across varying viewpoints while maintaining real-time rendering performance.

Despite these strides, significant hurdles remain in achieving robust generalization under sparse-view conditions and preserving temporal coherence during rapid motion. Current state-of-the-art techniques increasingly rely on hybrid representations that combine explicit Gaussian clouds

with implicit texture fields to enhance appearance fidelity and lighting consistency. By optimizing skinning weights and Gaussian parameters end-to-end through differentiable rendering, these models achieve superior expressiveness compared to purely implicit neural radiance fields, paving the way for high-fidelity digital avatars capable of nuanced emotional communication and dexterous manipulation in unbounded dynamic scenes.

3.2.3 Cloth Dynamics and Non-Rigid Articulation

Modeling cloth dynamics and non-rigid articulation presents unique challenges due to the high-frequency, topologically varying deformations that defy standard kinematic templates. Unlike rigid or purely articulated motion, cloth exhibits complex folding, stretching, and wrinkling patterns that require capturing both global pose changes and local soft-body physics. Recent advancements in dynamic 3D Gaussian Splatting address this by moving beyond per-frame respawning to establish persistent scene elements with identity, utilizing spatiotemporal encoders to track feature evolution across time while maintaining geometric consistency in rapidly changing regions [4].

To effectively regularize these deformations, state-of-the-art approaches integrate physics-based priors directly into the optimization framework. Methods such as GaussianBody leverage physical constraints within a canonical space to prevent artifacts in monocular video reconstruction, ensuring that cloth behaves plausibly under novel poses. Similarly, mesh-adsorbed frameworks attach Gaussian primitives to deformable surfaces, enabling the disentanglement of active simulation regions from static backgrounds. This hybrid representation allows for the application of As-Rigid-As-Possible losses and spring-mass systems, which enforce local rigidity where appropriate while permitting the necessary non-rigid flexibility for realistic fabric simulation.

Furthermore, handling unbounded dynamic scenes requires sophisticated decomposition strategies that distinguish between static structures and dynamic cloth regions based on motion gradients. By employing multi-level temporal hierarchies, these systems allocate higher-resolution primitives to areas of significant deformation, such as garment edges and folds, while reusing stable representations for static components. This

targeted allocation not only enhances rendering fidelity for fine details like texture and shadows but also facilitates real-time interaction and editing. Consequently, the integration of data-driven learning with explicit physical simulation marks a critical shift toward robust, generalizable models capable of synthesizing photorealistic non-rigid dynamics.

3.3 Semantic and Language-Embedded Scenes

3.3.1 Open-Vocabulary 3D Segmentation

Open-vocabulary 3D segmentation has emerged as a pivotal capability for enabling robots and autonomous systems to interpret unstructured environments without reliance on predefined category sets. Contemporary approaches predominantly leverage the synergy between 3D Gaussian Splatting (3DGS) and powerful 2D foundation models, such as the Segment Anything Model (SAM) and CLIP, to transfer semantic knowledge into three-dimensional space [14]. By lifting 2D semantic features or masks onto 3D Gaussian primitives, these methods facilitate query-based scene understanding, allowing for the identification and isolation of objects based on natural language prompts rather than fixed class labels.

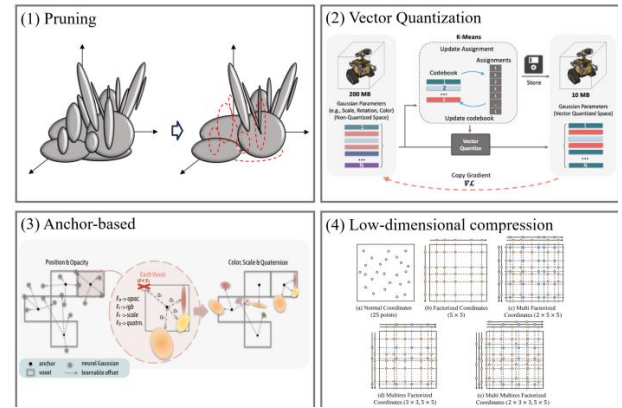


Figure 14: Chart from '3D Gaussian Splatting in Robotics: A Survey' (arXiv:2410.12262)

However, a significant technical challenge persists in ensuring semantic consistency across multiple views when relying solely on 2D priors. Since models like SAM guarantee coherence only within individual images, corresponding semantic regions often misalign when projected into 3D, leading to fragmented object representations and ambiguous boundaries. To mitigate this, recent frameworks have evolved from simple "splat-

then-segment" pipelines to more integrated strategies that enforce object-Gaussian correspondence during the reconstruction phase. Techniques such as embedding language features directly into Gaussian attributes or utilizing 3D-aware memory banks help regularize the semantic field, thereby reducing view-dependent artifacts and improving the geometric fidelity of segmented objects.

Furthermore, advanced methodologies are shifting towards unified frameworks that perform segmentation prior to or concurrently with reconstruction, effectively reversing the traditional pipeline to enhance object boundary precision. These approaches often incorporate spatial autocorrelation biases or conditional random fields within the optimization process to prevent overfitting, particularly in sparse-view scenarios where independent point primitives struggle. By jointly optimizing geometric and semantic properties, current state-of-the-art methods achieve robust open-world understanding capable of handling dynamic scenes and complex occlusions, marking a substantial advancement over earlier techniques that treated semantics as a post-processing step.

3.3.2 Multimodal Feature Distillation

Multimodal feature distillation has emerged as a pivotal strategy for enriching explicit 3D Gaussian representations with high-level semantic and perceptual information derived from 2D foundation models. By distilling features from pre-trained vision encoders, such as CLIP or DINO, into a compact 3D feature field, methods like Feature 3DGS enable open-vocabulary segmentation and language-guided scene editing without the computational overhead of per-ray neural network queries. This process typically involves optimizing Gaussian attributes to minimize the discrepancy between rendered feature maps and the corresponding 2D teacher features, effectively transferring rich semantic priors into the geometric primitive domain.

To address the inherent misalignment between 2D projections and 3D structures, advanced distillation frameworks incorporate spatial-temporal constraints and lightweight decoding architectures. Recent approaches utilize tri-plane neural fields or voxel grids as intermediate representations to model local texture fields in a shared manner, ensuring view consistency and temporal coherence in dynamic scenarios. Furthermore, techniques employing sparse optical flow and conditional ran-

dom fields refine dynamic labels, filtering incorrect associations and preventing the propagation of artifacts during motion. These mechanisms allow the distilled fields to maintain fidelity even under complex deformations, supporting applications ranging from robotic manipulation to interactive storytelling.

Despite these advances, challenges remain in balancing rendering throughput with feature expressiveness, particularly in 4D dynamic environments. While score distillation sampling effectively injects generative priors for novel view synthesis, extending these capabilities to time-varying scenes demands rigorous maintenance of physical consistency and motion dynamics. Current research focuses on decoupling static and dynamic components to optimize storage efficiency and training speed, often leveraging hierarchical feature pyramids to progressively refine Gaussian attributes. As the field evolves, the integration of multimodal distillation with generative diffusion models promises to further bridge the gap between semantic understanding and high-fidelity real-time rendering.

3.3.3 Text-Driven Scene Editing and Generation

Text-driven scene editing and generation have evolved from producing isolated 2D frame sequences to constructing comprehensive, volumetric 3D representations. Early approaches often lacked underlying geometric structure, failing to ensure temporal consistency in dynamic environments. Recent advancements leverage explicit representations like 3D Gaussian Splatting to overcome the computational inefficiencies and implicit limitations of Neural Radiance Fields [15]. By integrating pretrained diffusion models with rich semantic priors, these methods enable precise, prompt-based manipulation of scene elements while maintaining high-fidelity rendering and coherent multi-object interactions.

$$C(\vec{r}) = \sum_{i=1}^N \vec{c}_i \alpha_i \prod_{j=1}^{i-1} (1 - \alpha_j),$$

Figure 15: Chart from 'The Impact and Outlook of 3D Gaussian Splatting' (arXiv:2510.26694)

Current frameworks address the complexity of

scene composition by utilizing scene graphs and layout-guided generative processes. Methods such as CG3D interpret textual prompts as probabilistic graphical models to manage object relationships and physical realism, effectively mitigating artifacts like the Janus problem. Furthermore, techniques incorporating depth diffusion and inpainting allow for the seamless integration of new objects into existing environments. This compositional strategy ensures that generated scenes are not merely collections of assets but cohesive environments where lighting, geometry, and semantic alignment are optimized simultaneously through differentiable rendering pipelines.

The shift toward explicit Gaussian primitives has significantly enhanced the interactivity and scalability of text-driven workflows. Unlike implicit neural representations that require global optimization, explicit Gaussians facilitate localized editing, allowing users to modify specific attributes or animate components via custom visual editors without retraining the entire scene. This capability supports real-time feedback loops essential for robotic simulation and immersive content creation. Consequently, the field is transitioning from static reconstruction to dynamic, language-controllable systems capable of generating complex, animated 4D scenes that bridge the gap between human intent and machine-perceivable 3D structures.

3.4 Advanced Rendering and Physical Simulation

3.4.1 Inverse Rendering and Relighting

Inverse rendering within radiance fields aims to decompose observed imagery into intrinsic scene properties, including geometry, material reflectance, and illumination. While early Neural Radiance Fields implicitly encoded these factors, recent advancements leverage explicit 3D Gaussian Splatting to directly estimate physically based parameters such as albedo, roughness, and surface normals. By embedding Bidirectional Reflectance Distribution Function (BRDF) models into individual Gaussian primitives, these methods facilitate per-point shading calculations that surpass the limitations of view-dependent appearance modeling, enabling accurate separation of lighting from material characteristics.

Relighting capabilities are achieved by reconstructing high-dynamic-range environment maps

and integrating them with the decomposed material attributes. Techniques employing ray tracing against Gaussian distributions allow for realistic simulation of direct lighting, specular reflections, and soft shadows under novel illumination conditions. To address complex outdoor scenarios with significant illumination variance, some approaches incorporate compensation models or utilize diffusion priors to synthesize consistent multi-illumination data, ensuring that rendered outputs maintain photorealism even when light sources are drastically altered from the original capture conditions.

Despite these strides, challenges persist in handling dynamic scenes where motion complicates the consistent estimation of time-varying shadows and ambient occlusion. Standard optimization strategies often struggle with artifacts arising from inaccurate dynamic object segmentation or the interplay between static and moving elements under changing light. Consequently, current research focuses on refining temporal consistency through time-dependent shading functions and improving the robustness of geometry recovery in low-light or high-contrast environments to support applications in extended reality and virtual production.

3.4.2 Subsurface Scattering and Material Modeling

Standard 3D Gaussian Splatting typically relies on an emission-absorption model where lighting conditions are baked into spherical harmonics coefficients, inherently limiting the representation of complex light transport phenomena [16]. This formulation fails to account for subsurface scattering, a critical component for rendering translucent materials such as skin, wax, or marble, where light penetrates the surface, scatters internally, and exits at a different location. Consequently, naive splatting approaches often produce unrealistic results on such media, lacking the characteristic soft diffusion and color bleeding associated with volumetric interaction within the material body.

$$\mathbf{c}_i = \rho_i \odot \left(\sum_{m=1}^{(l+1)^2} \mathbf{L}_{env}^m \cdot \mathbf{O}_i^m \cdot \mathbf{d}_i^m \right),$$

Figure 16: Chart from 'Zero-Shot UAV Navigation in Forests via Relightable 3D Gaussian Splatting' (arXiv:2602.07101)

To address these limitations, recent advancements have integrated physical shading concepts directly into the Gaussian primitive definition. Methods like GaussianShader enrich splats with explicit material parameters, including diffuseness, specularity, roughness, and surface normals, moving beyond simple radiance field approximations. By incorporating classical reflection models and learnable radiance transfer functions, these approaches decouple intrinsic material properties from incident illumination. This separation enables accurate relighting and the simulation of specular highlights on reflective surfaces, effectively bridging the gap between efficient rasterization-based rendering and physically based shading pipelines.

Furthermore, specialized extensions have been developed to handle highly heterogeneous materials, such as human hair, which exhibit complex anisotropic scattering behaviors. By adapting volumetric ray-traced primitives or integrating convolutional encoders for driving signals, modern frameworks can capture fine geometric details and specific scattering lobes that standard isotropic Gaussians cannot represent. These innovations collectively enhance the fidelity of material modeling, allowing for the realistic synthesis of both opaque, reflective objects and translucent media while maintaining the real-time performance advantages inherent to the Gaussian splatting paradigm.

3.4.3 Physics-Constrained Deformable Objects

Physics-constrained deformable objects represent a significant evolution beyond purely data-driven deformation fields, integrating continuum mechanics directly into the 3D Gaussian Splatting framework. Unlike methods that rely solely on implicit neural networks to predict spatiotemporal displacements, these approaches treat individual Gaussians as discrete physical particles possessing volume, mass, and material properties. By embedding physics-based priors, such as elastoplasticity or spring-mass systems, directly into the optimization loop, these models ensure that reconstructed dynamics adhere to fundamental laws of motion, thereby mitigating temporal artifacts and geometric inconsistencies common in monocular dynamic reconstruction.

The core technical innovation lies in the unified simulation-to-rendering pipeline where the state

of each Gaussian is updated via physical solvers rather than learned deformation gradients alone. In this paradigm, the canonical Gaussian attributes are coupled with a mesh-based or particle-based simulation framework that computes forces and constraints at every time step. This allows for the accurate modeling of complex non-rigid behaviors, including stretching, compression, and topological variations, which are often challenging for standard kinematic models. The differentiable renderer then projects these physically simulated states into image space, enabling end-to-end optimization of both visual fidelity and physical plausibility from video observations.

Furthermore, recent advancements have extended these constraints to handle interactive scenarios and object manipulation by disentangling static backgrounds from actively simulated deformable entities. By enforcing conservation laws and material continuity, these systems achieve superior generalization to unseen viewpoints and temporal extrapolation compared to purely interpolative methods. The integration of explicit physical constraints not only enhances the realism of dynamic scene synthesis but also facilitates downstream applications in robotics and virtual reality, where accurate prediction of object behavior under external forces is critical for reliable interaction and control.

3.5 Domain-Specific Reconstructions

3.5.1 Underwater and Thermal Imaging

The extension of radiance field methodologies to underwater and thermal domains addresses unique physical constraints that degrade standard visual reconstruction. In underwater environments, light attenuation and wavelength-dependent scattering severely distort color fidelity and reduce contrast, challenging the photometric consistency assumptions inherent in Neural Radiance Fields (NeRF). Recent adaptations incorporate medium-specific volumetric rendering equations that explicitly model absorption and scattering coefficients, allowing the neural representation to disentangle scene geometry from water-induced degradation. These approaches often integrate physics-based priors to correct color casts and enhance visibility, enabling robust novel view synthesis where traditional Structure-from-Motion pipelines frequently fail due to feature scarcity.

Thermal imaging presents a distinct set of dif-

difficulties characterized by low spatial resolution, lack of texture, and ambiguous depth cues, which hinder accurate camera pose estimation and dense mapping. While visible-spectrum radiance fields rely on high-frequency textural details for optimization, thermal radiance fields must leverage temperature gradients and semantic boundaries to constrain the solution space. Emerging techniques fuse thermal data with complementary modalities, such as LiDAR or visible RGB, to provide geometric scaffolding for the implicit field. This multi-sensor fusion mitigates the ill-posed nature of monocular thermal reconstruction, facilitating the generation of coherent 3D representations essential for night-time surveillance and industrial inspection tasks.

Despite these advancements, significant hurdles remain in achieving real-time performance and generalization across varying environmental conditions. Underwater methods often suffer from high computational overhead due to the complexity of simulating light transport in participating media, while thermal approaches struggle with domain gaps when transferring pre-trained features from visible spectra. Current research trends indicate a shift toward efficient parameterizations, such as 3D Gaussian Splatting, to accelerate rendering speeds without sacrificing the ability to model complex physical phenomena [17]. Future developments are expected to focus on self-supervised learning strategies that reduce reliance on calibrated multi-sensor setups, thereby expanding the applicability of radiance fields in unstructured, extreme environments.

$$\mathcal{L}_{mc} = \sum_{n=1}^N |m_{n,t} - sg(m_{n,t-1})|.$$

Figure 17: Chart from 'Temporally Compressed 3D Gaussian Splatting for Dynamic Scenes' (arXiv:2412.05700)

3.5.2 Medical and Surgical Scene Modeling

Medical and surgical scene modeling presents unique challenges due to the highly deformable nature of soft tissues and the unbounded, topology-varying characteristics of endoscopic environments. Recent advancements leverage 3D Gaussian Splatting to reconstruct dynamic anatomical structures from monocular or sparse-view endo-

scopic videos, overcoming the limitations of traditional template-based approaches that struggle with non-rigid deformations [18]. By employing deformation fields to predict temporal updates for Gaussian parameters, these methods effectively capture complex tissue dynamics without relying on predefined kinematic trees or rigid motion assumptions.

$$\psi^x(t, \Theta^x) = \sum_{j=0}^{B-1} \omega_j^x \tilde{b}(t; \theta_j^x, \sigma_j^x) + \delta_x$$

Figure 18: Chart from 'ColorGS: High-fidelity Surgical Scene Reconstruction with Colored Gaussian Splatting' (arXiv:2508.18696)

A critical innovation in this domain involves the integration of depth ranking distillation and single-view dynamic reconstruction techniques to handle the lack of explicit camera poses and varying illumination typical in surgical settings. Unlike generic dynamic scene methods that often model deformation at the individual Gaussian level, emerging frameworks aim to encode semantically meaningful anatomical interactions by grouping Gaussians into instance-level representations [19]. This object-centric approach facilitates accurate isolation of surgical tools and organs, enabling robust tracking and mapping even under severe occlusions and rapid topological changes inherent to minimally invasive procedures.

$$\mathcal{L}_{cov} = \sum_{i=1}^N \sum_{k=1}^K \left\| d\left(\Sigma_c^{(i)}, \Sigma_c^{(k)}\right) - d\left(\Sigma_o^{(i)}, \Sigma_o^{(k)}\right) \right\|_1.$$

Figure 19: Chart from 'SurgicalGaussian: Deformable 3D Gaussians for High-Fidelity Surgical Scene Reconstruction' (arXiv:2407.05023)

Furthermore, the compression of large medical datasets into compact 3D Gaussian representations supports real-time, photorealistic rendering on lightweight extended reality devices, which is essential for intraoperative navigation and immersive surgical training [19]. By incorporating semantic segmentation and uncertainty-aware processing, these models enhance robustness against the low-texture and low-exposure conditions frequently encountered in endoscopy. Consequently, the shift towards interactive, high-fidelity 4D reconstructions not only improves preoperative diag-

nosis and planning but also lays the groundwork for next-generation augmented reality systems capable of providing precise, context-aware visual guidance during complex surgical interventions.

3.5.3 Large-Scale Urban and Aerial Mapping

Large-scale urban and aerial mapping presents unique scalability challenges due to vast spatial extents, complex motion patterns, and the need for high-fidelity reconstruction across city-level environments. Traditional radiance fields often struggle with prohibitive memory growth as scene size increases, necessitating divide-and-conquer strategies or hierarchical representations to manage computational loads. Recent advancements address these limitations by employing hybrid Gaussian representations and multi-plane encoders that efficiently capture both static infrastructure and dynamic elements like vehicles, enabling dense mapping suitable for autonomous navigation and urban planning.

To handle the intricacies of dynamic urban scenes, modern approaches integrate semantic information to decompose environments into static backgrounds and moving foregrounds. This semantic decomposition facilitates robust modeling of rapid motion and sparse camera setups, outperforming earlier hexplane-based methods in stability and accuracy. Furthermore, specialized techniques leverage object-level segmentation to localize dynamic entities, ensuring consistent novel view synthesis even under challenging conditions such as low-texture regions or varying illumination, which are common in extensive outdoor benchmarks.

Efficiency in reconstruction and localization remains critical for real-time applications, prompting the development of systems capable of on-line dense mapping and instant relocalization. By utilizing explicit 3D Gaussian splatting within voxel-based structures or KD-trees, these methods achieve rapid spatial queries and high-frame-rate rendering essential for robotics and augmented reality [20]. Current research directions focus on extending these capabilities to nighttime scenarios and reducing dependency on high-quality initial point clouds, aiming to create comprehensive, photorealistic digital twins that support lifelong operation in massive, unbounded environments.

$$L_{LiDAR}(\delta) = \frac{1}{s} \sum \|P(G_{comp}) - L_s\|^2$$

Figure 20: Chart from 'DrivingGaussian: Composite Gaussian Splatting for Surrounding Dynamic Autonomous Driving Scenes' (arXiv:2312.07920)

4 Neural Radiance Fields for Dynamic Scene Synthesis

4.1 Implicit Dynamic Representations

4.1.1 Canonical Volume Warping

Canonical volume warping establishes a foundational paradigm for dynamic scene reconstruction by mapping time-varying observations into a static, canonical coordinate space. This approach decouples geometry and appearance from temporal deformation, allowing the scene to be represented as a single volumetric field that remains invariant across time steps. By learning a deformation field, typically parameterized by neural networks or explicit grids, methods transform query points from observation space back to this canonical volume before evaluating density and radiance. This formulation effectively reduces the complex problem of 4D reconstruction to a more tractable 3D representation learning task, leveraging the robustness of static neural radiance fields while accommodating non-rigid motion.

The core mechanism relies on bidirectional mapping functions that warp spatial coordinates based on temporal embeddings or physical priors. During rendering, rays cast from camera viewpoints are sampled, and each sample point is inversely warped into the canonical frame using the learned deformation model. The volumetric properties are then queried from the static canonical field, ensuring temporal consistency and reducing the memory footprint compared to storing independent volumes for every time step. Advanced variants integrate graph neural networks or attention mechanisms to bias deformations toward local body parts, thereby mitigating spurious correlations and improving the fidelity of articulated movements in highly deformable actors.

Despite its effectiveness, canonical warping faces challenges in handling topological changes and extreme deformations where the bijective mapping assumption breaks down. Recent advancements address these limitations by combining canonical representations with explicit point-

based primitives or hybrid grid structures to better capture high-frequency details and discontinuous motions. Furthermore, integration with transformer architectures enables more robust aggregation of multi-view features along warped rays, enhancing generalization to unseen poses. While early methods struggled with over-smoothing in complex scenes, modern iterations employ distortion losses and metric-aware deformation fields to preserve sharp geometric boundaries and maintain photorealistic rendering quality under significant non-rigid transformations.

4.1.2 Neural Motion Factorization

Neural motion factorization addresses the inherent complexity of dynamic scenes by decomposing spatiotemporal representations into distinct static backgrounds, rigid object trajectories, and non-rigid deformation fields [21]. Unlike monolithic 4D neural fields that struggle to disentangle geometry from motion, this paradigm explicitly models scene dynamics through separable latent components. By isolating motion parameters from appearance features, these methods effectively mitigate artifacts caused by occlusions and topological changes, enabling robust reconstruction of complex interactions such as human articulation within cluttered environments.

$$\phi_{\mathbf{x}} = \sum_{n=1}^{N_g} G_n(\mathbf{x} | \rho_n, \mu_n, \Sigma_n)$$

Figure 21: Chart from 'DIGS: Dynamic CBCT Reconstruction using Deformation-Informed 4D Gaussian Splatting and a Low-Rank Free-Form Deformation Model' (arXiv:2506.22280)

Recent advancements leverage object-centric neural scattering functions and decomposed radiance fields to achieve this separation without relying on predefined motion priors or object templates. Techniques often employ joint optimization frameworks that simultaneously learn neural representations, object tracking, and volume rendering. To ensure physical plausibility, these approaches integrate specific regularization terms, such as temporal smoothness constraints on scene flow and consistency losses between forward and backward motion fields. This structured decomposition allows the network to distinguish between

camera ego-motion and independent object movement, significantly improving the fidelity of novel view synthesis in highly dynamic sequences.

The integration of efficient explicit representations, such as 3D Gaussian splatting, has further accelerated neural motion factorization, enabling real-time rendering and interactive control [21]. By factorizing motion into low-dimensional trajectory parameters and high-frequency deformation residuals, these models achieve superior generalization to unseen motion patterns while maintaining sharp geometric details. This modular architecture not only enhances visual quality by preserving high-frequency textures during rapid movement but also facilitates downstream applications like zero-shot moving object segmentation and precise 3D point tracking, establishing a versatile foundation for comprehensive dynamic scene understanding.

4.1.3 Continuous Space-Time Dynamics

Continuous space-time dynamics move beyond discrete frame-to-frame transformations by modeling scene evolution as a differentiable function over a unified four-dimensional domain. Unlike time-global representations that assume static geometry warping or purely local models treating frames independently, continuous approaches leverage Neural Ordinary Differential Equations to define smooth trajectories. This formulation ensures invertibility and physical plausibility, often incorporating Hamiltonian mechanics to enforce energy conservation and system stability throughout extended temporal sequences without the drift associated with standard Euler integration.

To achieve high-fidelity rendering at arbitrary, non-observed timestamps, these methods employ sophisticated interpolation strategies within a canonical space. By decomposing motion into backward and forward temporal components relative to neighboring observations, the system calculates precise deformation fields using time-variant blending coefficients. This allows for the synthesis of novel views at continuous target times $\tilde{t} \in \mathbb{R}$, effectively bridging gaps between discrete captures. The integration of symplectic solvers further guarantees that the latent route maintains motion continuity, enabling robust reconstruction of both rigid and non-rigid objects under complex dynamic conditions.

Advanced implementations frequently adopt spatial-temporal decomposition to optimize com-

putational efficiency and representational capacity. By distinguishing between static backgrounds and dynamic foregrounds, these models apply selective filtering where only relevant spatial locations undergo deformation at specific moments. Techniques such as voxel-level variation fields and dynamic neural point clouds facilitate this dual selectivity, ensuring that the learned dynamics generalize across unseen initial states. Consequently, the resulting representations support downstream tasks like model-predictive control and interactive novel view synthesis, providing a rigorous framework for understanding light transport and geometric evolution in continuously changing environments.

4.2 Hybrid Explicit-Implicit Frameworks

4.2.1 Voxel-Grid Temporal Encoding

Voxel-grid temporal encoding extends traditional volumetric representations to dynamic scenes by integrating time as an explicit dimension within the grid structure. While standard voxel grids discretize space into fixed-size cells containing geometric or appearance features, dynamic formulations augment this with a temporal axis, creating 4D tensors that capture scene evolution. Early approaches often treated time as an additional coordinate for trilinear interpolation, yet this naive extension exacerbates the inherent cubic memory growth, rendering high-resolution spatiotemporal reconstruction computationally prohibitive without specialized sparsity mechanisms.

To mitigate these memory constraints, recent methodologies employ sparse voxel grids coupled with time-aware neural primitives. Instead of storing dense 4D volumes, these techniques utilize learnable feature vectors at sparse spatial locations, where temporal dynamics are modeled through compact neural networks or low-rank decompositions. A prominent strategy involves decomposing the 4D encoding into separate spatial and temporal components, such as using multi-resolution hash encodings or factorized tensor representations. This allows the system to query time-variant features efficiently by combining static spatial embeddings with dynamic temporal codes, significantly reducing the parameter count while preserving fine-grained motion details.

Furthermore, advanced encoding schemes incorporate inner-product time queries and sinusoidal positional embeddings to enhance temporal

coherence. By concatenating time tokens with spatial features before feeding them into lightweight multilayer perceptrons, these models effectively disentangle static geometry from dynamic deformation. Such architectures enable fast training and real-time rendering of complex dynamic scenes, overcoming the limitations of purely implicit representations. The integration of sparse data structures with explicit temporal modeling thus represents a critical advancement, balancing the trade-off between representation capacity and computational efficiency in 4D novel view synthesis.

4.2.2 Point-Based Neural Primitives

Point-based neural primitives address the controllability limitations inherent in pure neural field representations by discretizing scene geometry into learnable point clouds. Unlike continuous implicit functions that require dense sampling, these methods associate latent feature vectors with explicit 3D coordinates, enabling direct manipulation of geometric primitives similar to traditional mesh editing. This hybrid approach leverages the high-frequency detail capture of neural networks while maintaining the structural interpretability of point clouds, facilitating efficient rendering through localized feature querying rather than global network evaluation.

To enhance generalization across object categories, recent advancements incorporate latent class encodings that condition the neural representation on specific shape descriptors. By injecting a learned latent vector into the input query alongside spatial coordinates, a single shared network can model distributions of similar objects, effectively decoupling geometry from appearance attributes. Furthermore, techniques utilizing multi-resolution hash encodings have significantly accelerated training and inference by replacing deep fully connected layers with lightweight lookups, allowing sparse point sets to represent complex scenes with reduced computational overhead and memory footprint.

In dynamic scenarios, point-based primitives are extended through spatiotemporal formulations that distinguish between static background elements and deformable foreground objects. Methods employ template-free articulated models or motion-guided deformation fields to update point positions and features over time without requiring predefined skeletal structures. To prevent over-parameterization and artifacts in stable regions,

regularization strategies such as L1 penalties on motion coefficients are applied, ensuring that only necessary primitives undergo dynamic transformation. This selective modeling preserves rendering efficiency while accurately capturing high-frequency motion details in non-rigid sequences.

4.2.3 Mesh-Guided Neural Deformation

Mesh-guided neural deformation integrates explicit topological priors with implicit neural representations to address the ill-posed nature of non-rigid reconstruction. By anchoring deformation fields to a coarse template mesh, these methods leverage established techniques like embedded deformation graphs to enforce structural coherence. This hybrid approach restricts the solution space to physically plausible configurations, effectively preventing topological inconsistencies that often plague purely data-driven implicit models during extreme articulation or occlusion.

The core mechanism typically involves mapping canonical neural features to deformed space through learnable transformations derived from mesh vertices. Instead of predicting dense voxel displacements directly, the network regresses node transformations that influence surrounding neural primitives via smooth interpolation weights. This formulation significantly reduces the number of free parameters required to model complex dynamics, accelerating convergence while ensuring that local geometric details remain consistent with the global motion trajectory defined by the underlying skeleton or mesh topology.

Recent advancements have extended this paradigm to handle highly deformable scenes by incorporating bijective constraints and physics-informed regularization. These enhancements ensure that the deformation field remains invertible, preserving volume and preventing self-intersections during synthesis. Furthermore, decoupling the optimization of mesh-driven global motion from neural refinement of local appearance allows for stable training dynamics. Consequently, mesh-guided strategies currently represent a robust frontier for achieving high-fidelity novel view synthesis in dynamic environments where pure implicit methods struggle with temporal stability.

4.3 Generalizable and Data-Efficient Synthesis

4.3.1 Zero-Shot Novel View Synthesis

Zero-shot novel view synthesis addresses the critical challenge of rendering unseen perspectives from dynamic scenes without requiring per-scene optimization or dense multi-view supervision. Unlike traditional Neural Radiance Fields that rely on extensive gradient descent to fit specific scene geometries, zero-shot approaches leverage pre-trained generalizable priors to infer 3D structure and appearance from sparse inputs instantly. This paradigm shift enables real-time applications by decoupling the heavy computational cost of training from the inference phase, allowing models to generalize across diverse, previously unobserved environments including non-rigidly deforming objects.

To achieve robust generalization, contemporary methods often integrate explicit motion modeling with efficient rendering backbones such as Multi-Plane Images or 3D Gaussian Splatting [4]. A prevalent strategy involves combining a frozen, generalizable view synthesis network with a learnable scene flow field that captures temporal deformations specific to the test sequence. By disentangling static background geometry from dynamic foreground motion through factorized motion fields, these systems can accurately handle complex occlusions and non-rigid transformations. The scene flow acts as a deformation module that warps a canonical representation into time-varying states, ensuring spatiotemporal consistency even when input views are limited to monocular video streams.

Despite significant progress, achieving high-fidelity synthesis in zero-shot settings remains constrained by the difficulty of disambiguating depth and motion from sparse observations. Current research focuses on enhancing geometric priors through spherical-BEV projections and iterative hypothesis generation to better constrain the solution space for dynamic environments. Furthermore, optimizing the trade-off between rendering speed and visual quality is paramount, with recent advancements demonstrating the ability to synthesize photorealistic frames at hundreds of frames per second. Future directions aim to improve the stability of pose-free synthesis and extend these capabilities to handle varying illumination conditions and complex object interactions without re-

training.

4.3.2 Recurrent Latent Rendering Modules

Recurrent latent rendering modules address the challenge of modeling temporal coherence in dynamic scenes by evolving a compact latent state over time rather than storing independent representations for each frame. These architectures typically employ recurrent neural networks or ordinary differential equations to propagate a view-agnostic latent code, which captures the underlying scene dynamics and deformation fields. By conditioning the neural radiance field on this evolving latent state, the system effectively decouples static geometry from time-varying motion, allowing for continuous trajectory extrapolation and consistent novel view synthesis even under sparse observation densities.

The core mechanism involves updating the latent representation at each time step based on previous states and current inputs, often utilizing iterative refinement strategies to handle complex non-rigid deformations and occlusions. This approach enables the network to learn a shared manifold for object appearances and motions, significantly reducing memory overhead compared to per-frame optimization. Advanced implementations integrate motion-aware bundle adjustment or variational autoencoders to regularize the latent space, ensuring that the predicted future states remain physically plausible while maintaining high-fidelity rendering quality across long temporal horizons.

Furthermore, these modules facilitate the separation of global illumination effects from individual object radiance fields by leveraging factored representations within the recurrent loop. By freezing background components and only updating latent codes associated with dynamic entities, the rendering pipeline achieves efficient convergence and robustness against transient artifacts. This design supports flexible manipulation of object trajectories and placements post-training, as the learned latent dynamics can be queried at arbitrary time intervals to synthesize coherent video sequences without re-optimizing the entire scene representation.

4.3.3 Cost-Volume Based Depth Estimation

Cost-volume based depth estimation constructs a 3D cost volume by sampling features along epipolar lines across multiple views, effectively encod-

ing matching costs for hypothetical depth planes. This representation leverages epipolar geometry to constrain the search space, where the baseline and focal length dictate the discretized depth range. By aggregating feature correlations from stereo pairs or multi-view sequences, the method transforms the depth regression problem into a classification task over discrete depth bins, enabling robust geometric reasoning without explicit 3D supervision.

To derive final depth maps, the constructed cost volume undergoes regularization and aggregation via 3D convolutions or attention mechanisms that enforce spatial consistency and smoothness. Advanced variants integrate learnable modules to refine coarse cost volumes, handling textureless regions and occlusions that traditionally plague passive stereo methods. The integration of ray-based attention further enhances this process by weighting sampled points along camera rays, ensuring that the estimated depth corresponds to solid surfaces rather than transparent or ambiguous volumetric regions.

Recent developments extend this paradigm to dynamic scenes and zero-shot transfer scenarios by incorporating pseudo-depth priors from monocular estimators. These hybrid approaches utilize pre-trained networks to initialize cost volumes or impose geometric constraints, significantly improving generalization across diverse datasets. Furthermore, differentiable volumetric rendering techniques allow for end-to-end optimization of the cost volume construction, bridging the gap between traditional multi-view stereo and neural implicit representations for high-fidelity scene reconstruction [22].

$$\mathbf{r}_i^D(t) = \sum_{k=0}^{n_r} r_{i,k} (t - \mu_i^\tau)^k$$

Figure 22: Chart from 'DSD-GS: Dynamic-Static Decomposition of Gaussian Splatting for Efficient and High-Fidelity Dynamic Scene Reconstruction' (arXiv:2605.30863)

The core advantage of this architecture lies in its capacity for editable and compositional synthesis. By treating objects as reusable modules within the graph, the framework supports high-level semantic operations such as object insertion, removal, duplication, and trajectory modification. The learned scene flow or transformation matrices associated with graph edges facilitate the coherent movement of objects through time, ensuring that novel configurations maintain physical plausibility and consistent lighting interactions. Consequently, users can construct entirely new scene graphs by sampling from the learned distribution of objects and transformations, generating photo-realistic views of scenarios never observed during

training.

Despite its flexibility, scaling this approach to crowded scenes remains computationally demanding, as it requires optimizing distinct neural primitives for every entity. Furthermore, maintaining global illumination consistency across dynamically rearranged objects poses significant challenges compared to static settings. Nevertheless, scene graph-based methods represent a critical shift towards semantically aware dynamic rendering, bridging the gap between low-level pixel synthesis and high-level scene understanding. This structured representation is particularly vital for applications requiring interactive control, such as virtual reality and robotic simulation, where precise manipulation of dynamic elements is paramount.

4.4.3 Inter-Object Shadow and Light Transport

Accurate modeling of inter-object shadow and light transport remains a significant challenge in dynamic scene reconstruction, particularly when objects undergo motion that alters incoming radiance directions. Traditional baking techniques fail under these conditions as they assume static illumination, necessitating approaches that explicitly simulate physical light transport and environment maps to support relighting and scene editing. While many non-rigid view synthesis methods rely on simplified emission-absorption models that encode global illumination cues implicitly within surface colors, these approaches often struggle to disentangle spatial and temporal deformations, leading to artifacts when objects interact or occlude one another.

To address these limitations, advanced frameworks incorporate volumetric path tracing and shadow ray casting to compute occlusions and indirect illumination dynamically. By evaluating transmittance along rays and masking incident light with occlusion maps, these methods can synthesize hard cast shadows and specular reflections that respond correctly to moving light sources and geometry. Unlike standard neural radiance field implementations that render in a single forward pass, sophisticated architectures project camera rays across multiple observations or utilize layered Gaussian residuals to resolve complex diffractions and contact shadows, ensuring that specular highlights move consistently with fixed scene illumination even as viewpoints change [1].

Despite progress, fully differentiable light transport in dynamic settings often requires balancing computational cost with physical fidelity. Current solutions increasingly adopt hybrid strategies, combining explicit geometric representations with learned heuristics to approximate inter-object lighting effects without retraining for every new configuration. These methods enable the synthesis of realistic shadows and reflections for moving entities, allowing for robust scene composition and object extraction. However, achieving real-time performance while maintaining accuracy in scenarios involving complex human-scene interactions and irregular point cloud structures continues to drive research toward more efficient explicit radiance fields and adaptive sampling techniques.

5 Future Directions

Despite the significant strides made in dynamic 3D Gaussian Splatting, current methodologies still face substantial limitations that hinder their widespread deployment in unconstrained real-world scenarios. A primary bottleneck remains the struggle to robustly handle drastic topological changes, such as objects appearing, disappearing, or undergoing severe structural fragmentation, which often leads to optimization instability and temporal flickering. Furthermore, existing approaches frequently rely on heavy supervision from external priors, including pre-computed optical flow, depth maps, or semantic segmentation masks, making them less adaptable to sparse-view settings or environments where such auxiliary data is unavailable or noisy. The trade-off between compression efficiency and rendering fidelity also persists; while recent pruning and quantization techniques reduce memory footprints, they often introduce visible artifacts during rapid motion or fail to maintain consistent quality over extended temporal sequences. Finally, the physical plausibility of generated motions is often superficial, lacking true adherence to physical laws, which limits the utility of these reconstructions for downstream tasks like robotic interaction or physics-based simulation.

Future research should prioritize the development of fully self-supervised frameworks that eliminate the dependency on external 2D or 3D priors, enabling robust reconstruction from casual, uncalibrated video captures. This involves designing novel loss functions and architectural

inductive biases that can implicitly disentangle static and dynamic components while simultaneously inferring camera motion and scene geometry. Another critical direction is the integration of generative priors and diffusion models to hallucinate missing geometric details and enforce temporal consistency in occluded regions, thereby addressing the ill-posed nature of dynamic reconstruction under sparse observations. Additionally, there is a pressing need to embed explicit physical constraints directly into the optimization process, moving beyond simple smoothness regularizers to incorporate rigid body dynamics, collision avoidance, and material properties. Such physics-informed neural rendering would not only stabilize training but also ensure that the reconstructed scenes behave realistically under external forces. Finally, advancing adaptive compression schemes that dynamically adjust the density and precision of Gaussian primitives based on motion complexity and viewer attention could drastically improve scalability without compromising visual quality.

The successful realization of these future directions promises to transform dynamic 3D reconstruction from a specialized research tool into a ubiquitous technology capable of operating in real-time on edge devices. By achieving robust, prior-free reconstruction with high physical fidelity, these advancements will unlock new possibilities in autonomous navigation, allowing robots to interact safely with highly deformable objects and unpredictable environments. In the realm of virtual and augmented reality, improved compression and temporal stability will facilitate the streaming of immersive, interactive 4D experiences over standard bandwidths, revolutionizing telepresence and digital entertainment. Moreover, the ability to accurately model complex physical interactions will bridge the gap between computer vision and robotics simulation, providing high-fidelity digital twins for training agents in safety-critical applications. Ultimately, overcoming current limitations will establish a new standard for capturing and representing the dynamic nature of our world, fostering innovation across diverse sectors ranging from healthcare to heritage preservation.

6 Conclusion

This survey has provided a comprehensive examination of the rapidly evolving landscape of dynamic 3D scene reconstruction using Gaussian

Splatting. We systematically categorized existing methodologies into three primary domains: spatiotemporal motion modeling, dynamic-static scene decomposition, and efficient representation strategies. Our analysis reveals that while canonical space deformation fields offer a compact basis for handling non-rigid transformations, they often struggle with topological changes and large displacements. Hierarchical motion anchors and physics-informed trajectory predictions have emerged as robust alternatives, effectively decoupling geometry from motion to stabilize optimization and capture high-frequency details. Furthermore, we highlighted the critical shift from relying on external 2D priors to learning adaptive 3D masks and rigidity-aware constraints, which significantly mitigate artifacts such as temporal jitter and boundary leakage. Finally, the review underscored the necessity of sparse parameterization and temporal pruning to achieve the storage efficiency required for practical deployment without compromising the real-time rendering capabilities that define the Gaussian Splatting paradigm.

The significance of this work lies in its synthesis of disparate research threads into a unified taxonomy, offering researchers a clear perspective on the trade-offs between fidelity, efficiency, and physical plausibility in 4D reconstruction. By critically evaluating the strengths and limitations of current approaches, this survey identifies fundamental bottlenecks that have hindered the widespread adoption of dynamic Gaussian Splatting in interactive applications. It elucidates how the transition from implicit neural representations to explicit Gaussian primitives has revolutionized the field, enabling photorealistic novel view synthesis at frame rates previously unattainable. Moreover, by dissecting the mechanisms behind successful motion decoupling and compression, this review provides a foundational reference for developing next-generation algorithms capable of handling complex, unconstrained environments. It serves as an essential guide for navigating the methodological choices required to balance computational overhead with the demands of high-quality dynamic scene capture.

Looking forward, several promising avenues remain to be explored to fully realize the potential of dynamic Gaussian Splatting. Future research must prioritize the integration of more robust physical priors to handle extreme topological changes and complex fluid dynamics, which re-

main challenging for current deformation models. There is also an urgent need for advanced compression techniques that can further reduce memory footprints to facilitate deployment on resource-constrained edge devices and mobile platforms. Additionally, the community should focus on developing self-supervised learning frameworks that minimize reliance on accurate camera poses or external segmentation masks, thereby enhancing robustness in wild, uncontrolled settings. We call upon researchers to bridge the gap between theoretical innovation and practical utility, fostering collaborations that push the boundaries of real-time 4D reconstruction. By addressing these open challenges, the field can move beyond static demonstrations toward pervasive applications in autonomous navigation, immersive virtual reality, and digital heritage preservation.

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